

Disclaimer: These slides must be used in combination with all resources provided on STEMLearn. If anything is unclear, please consult me.

MNS Final Year Project: Project Proposals

Due: 13 March @ 17h00
(12 month) but willing to
accept it on 16 March



Example*: Manufacture of an internally cooled blade for use in gas turbines



Source: www.digitalengineering247.com



Source: www.eos.info

*Used as a demonstrative example throughout these slides

Spirit of a project proposal (part 1)

NOT just a PAPER EXERCISE

→ Serves a very early draft for your final submission

→ Also good for PLANNING

Well begun
is half
done.

Aristotle

www.quote-coyote.com



John Keats



There is an old saying "well begun is half done"-'tis a bad one. I would use instead-Not begun at all 'til half done.

AZ QUOTES

Spirit of a project proposal (part 2)

1. Establish that **the need exists**: Introduction and Motivation
 - I. Current designs are unable to meet high temperature and pressure requirements (Denver et al. 2018)
 - II. Although materials and designs are available to manufacture suitable blades, the process is expensive and has long lead times (Johnson et al. 2019)
 - III. Traditional manufacturing processes are unable to reproduce the intricate internal cooling channels necessary for this design (Liney et al. 2020)
2. Highlight and prove that the **solution you propose is right**: Whole document
 - I. Introduction: Additive manufacturing (AM) is a breakthrough technology that overcomes the geometrical limitations typically associated with traditional manufacturing routes. Furthermore, selective laser melting (SLM) is an ideal AM technology for the manufacturing of complex shapes (such as internally channelled turbine blades) from nickel-base superalloys (Linthon et al. 2015; Bunta et al. 2019)
 - II. Objectives/Aims: Highlights the thinking behind the workplan for the solution. E.g.
 1. Aim: To manufacture an internally cooled blade using SLM
 2. Objective 1: Establish the engineering specifications for the turbine blade
 3. Objective 2: Determine the appropriate materials based on requirements
 4. Objective 3: Determine SLM processing parameters (laser scan speed, powder, etc.)
 5. and so forth...

Spirit of a project proposal (part 3)

3. Show that you are **capable of completing** the project with techno-economic analysis:
 - I. Show budget is within reasonable limits (remember the R5000 running cost limit)
 - II. Show timeline is achievable (take note of module framework deadlines)
4. End of life and resource use considerations
5. ECSA self-assessment

- For this and all subsequent reports:
- Follow “Writing Guide” on STEMLearn
 - Will not correct writing errors more than once:
more time to focus on technical content.

To start: Executive Summary

Executive Summary	
Title of Project	
Objectives	
What is current practice and what are its limitations?	
What is new in this project?	
If the project is successful, how will it make a difference?	
What are the risks to the project being a success? Why is it expected to be successful?	
What contributions have/will other students made/make?	
Which aspects of the project will carry on after completion and why?	
What arrangements have been/will be made to expedite continuation?	

Template is available
on STEMLearn

Include at least 5 references

Introduction (AKA Background)

Broad to narrow

Eg broad: Aerospace safety is of utmost importance, especially in commercial fields.

Turbine blades must withstand high temperatures of over 1000 °C, requiring improved cooling methods in a complex design.

AM/SLM offers a suitable alternative to standard manufacturing processes as it allows the manufacturing of complex designs at lower cost and faster lead times.

There are, however, many unknowns associated with SLM of nickel superalloys, such as processing parameters and fatigue strength, as the process is known to cause porosity issues.

This project aims to solve this issue by investigating the design and production of an internally cooled turbine blade using SLM of nickel superalloys.

This work forms part of Ms KD Larkey's final year Mechanical Engineering 478 Project. It will take place from 2 January until 24 October. Experimental facilities within the STC-LAM and Structure's laboratory at SU will be used within a budget of R5000.

What

Why

How

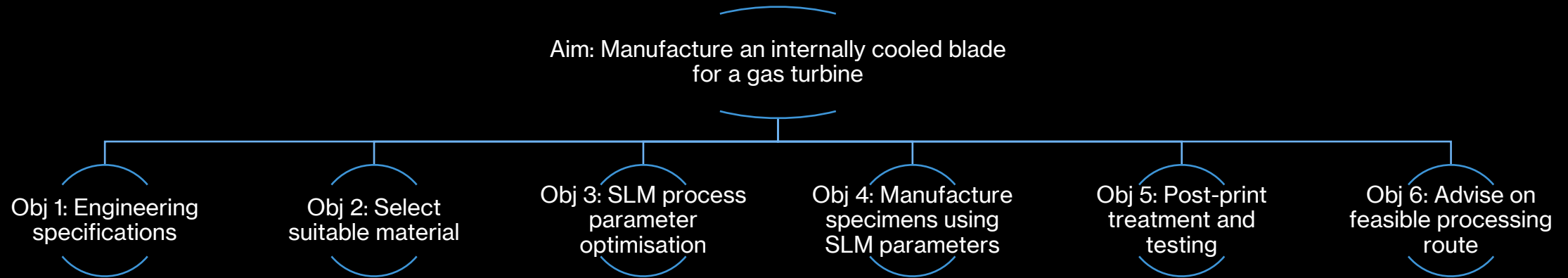
Who,
Where,
When,
How
much?

Introduction (part 2)

- **Introduce terms/concepts** so that the objectives and motivation sections that follow the introduction make sense. E.g. Definitions of additive manufacturing or selective laser melting in this example.
- Refer to **previous works** if your project is a continuation. E.g. A previous student (Neaves, 2025) developed the powder processing methods that will be employed in this work.
 - Be sure to highlight in your Introduction how YOUR work differs/improves on previous work.
- End with **a final statement** summarising the parameters of the project. E.g.
 - “This work forms part of Ms KD Larkey’s final year Mechanical Engineering 478 Project. It will take place from 9 February until 23 October. Experimental facilities within the Metallography Laboratory and Structures Laboratory at Stellenbosch University will be used within a budget of R5 000.”

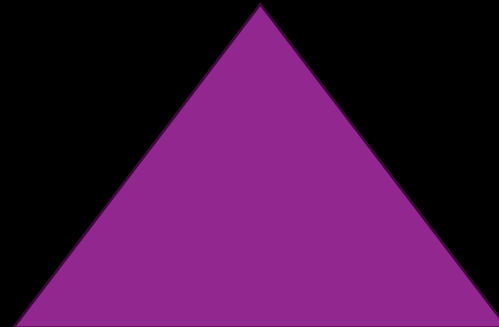
Objectives

- About 4-6 NUMBERED objectives
- Clear and concise
- Use action words where possible
- Measurable: compare Introduction and Conclusion
- Can you answer the question: Have you met all your objectives?
- Brief and client (Dr Melody Neaves) is a good resource to formulate these
- Can include scope here: limits of the study.
- Usually link objectives to a single aim:

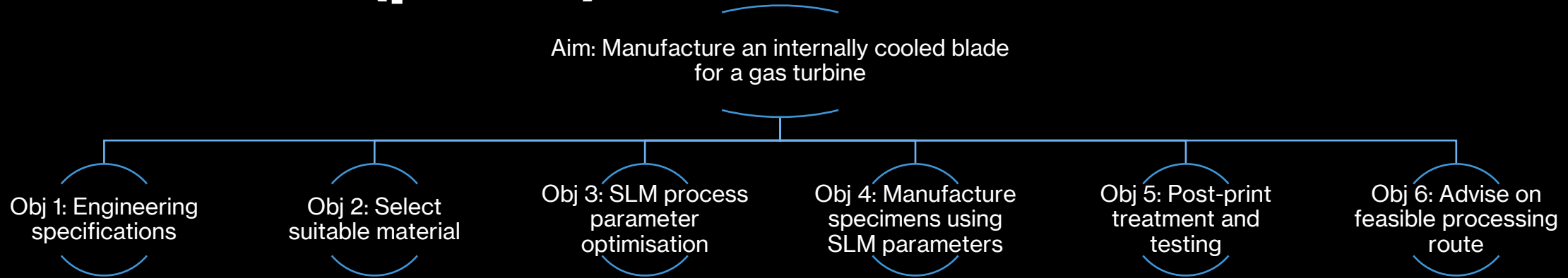


Motivation (part 1)

- Just answer in half a page to a page the following question:
- Why is YOUR SPECIFIC solution and OBJECTIVES/AIM a suitable solution?
- Do NOT just justify background of study: Already did this in Introduction.
- Expand on why your solution is competitive and why it should be invested in.
- Include references here, where necessary.
- Use the upside down (or reverse) funnel method.



Motivation (part 2)



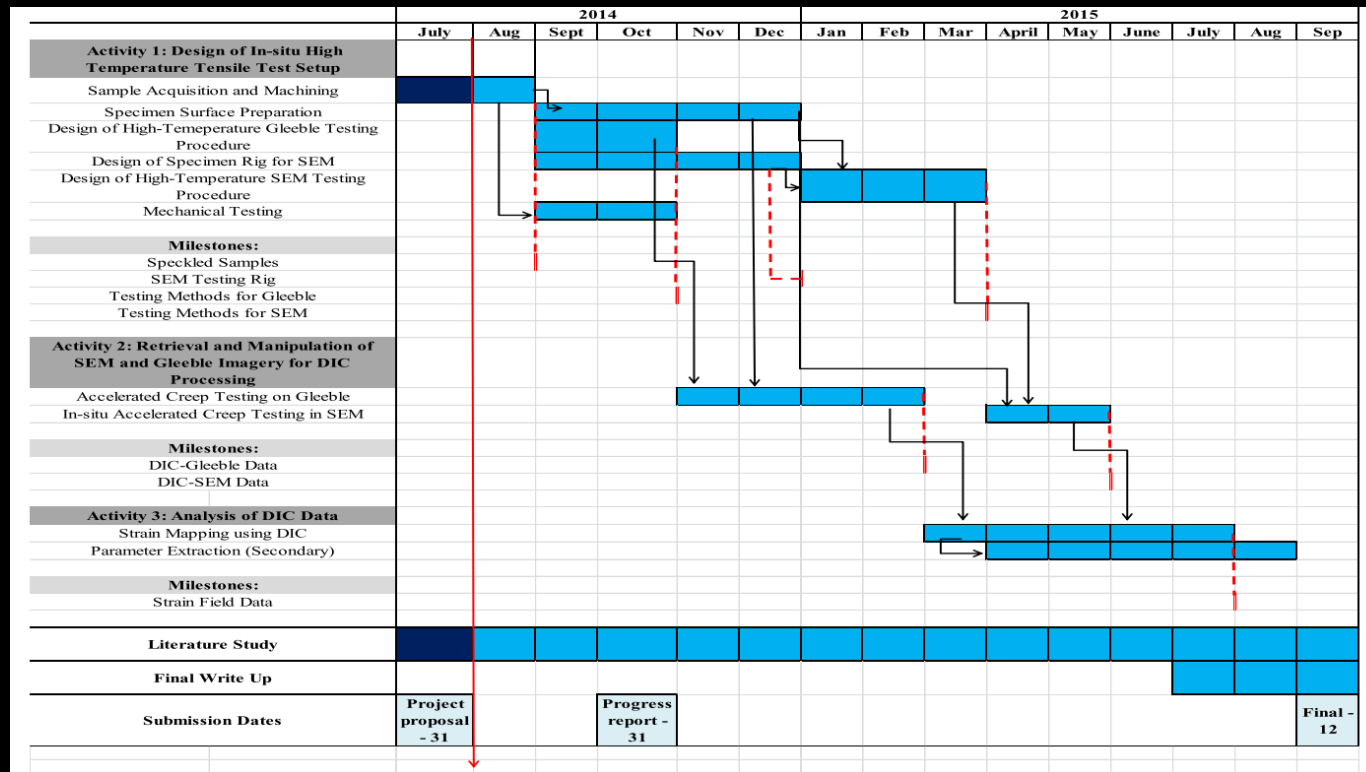
- Take the aim (manufacture of an internally cooled blade for a gas turbine...) and make that the **narrow part of the funnel**: the immediate output/benefit, such as what do you end up with after you complete all the objectives, ie. the overall aim. E.g.
 - “By successfully establishing a verified SLM processing and post-processing route, this project provides a validated framework for manufacturing complex internally cooled blade designs that are difficult to produce using conventional techniques. ”
- Now move to a “**medium**” benefit/impact such as industry relevance. E.g.
 - “This defined processing route removes the trial-and-error bottleneck for turbine manufacturers, enabling them to adopt additive manufacturing to produce advanced cooling channel designs.”
- Finally the “**broadest**” benefit/impact such as societal/global/research impact. E.g.
 - “Ultimately, facilitating the reliable production of these complex, internally cooled blades allows for higher turbine operating temperatures, which directly translates to increased thermal efficiency, reduced fuel consumption as well as lower carbon emissions across the aerospace and energy generation sectors.”

Planned Activities

- About 10 – 12 activities
- Be very specific to your project:
 - Not specific e.g.: Manufacture design
 - Specific e.g.: Produce standardised tensile specimens on SLM machine using optimised process parameters
- Use these activities in your Gantt chart
- E.g.
- Activity 1: Research viable materials and optimum processing parameters using an SLM machine
 - Conduct a literature review on aerospace-suited materials, including Ni superalloys. Subsequently establish optimum processing parameters based on literature review of SLM-produced materials.
- Activity 2: Produce tensile specimens on SLM machine using optimised process parameters
 - Use optimised laser power, scan speed, hatch spacing and scan strategy parameters to manufacture tensile specimens on the SLM machine.
-
- Activity 11: Summarise findings regarding SLM-printed blade and report to aerospace client
 - Make recommendations regarding the most suitable processing route using SLM machines for the production of dense, strong and fatigue-resistant turbine blades. Summarise findings in a technical report.

Gantt Chart

- Can use MS Excel and export as a (good quality) image (e.g. see below)
- Indicate milestones: important dates (see study guide) for reports and presentations, project milestones such as initial design review. E.g. Optimum process parameter database is one milestone, first blade prototype build is another
- Parallel vs sequential: think carefully about this
- Allow 2-3 weeks between submitting drawings to workshop and start of work, 4-6 weeks for delivery of specialised or imported components
- See STEMLearn slides for more guidance



Budget

- Refer to **STEMLearn!**
- Different for each project
- Replacement value for common machines:
 - DIC (commercial hardware and software): R1.2 million
 - Tensile tester: R1 000 000
 - Fatigue tester: R1 200 000
 - Microscope (stereo/optical): R350 000
 - Polishing/grinding metallography equipment: R250 000
 - Ask me or laboratory technicians for other specialised equipment OR look for approximate value online

Project Risk Assessment

- Risks that prevent the project from being successful
- Must also provide solutions and/or alternatives.
- E.g.
 - **Budget** (give specific risk items, for e.g. the SLM powder may be too expensive)
 - **Time** (give specific risk timelines, for e.g. load shedding may delay the specimen printing process which would delay all subsequent activities)
 - **Technical difficulties:** For instance, perhaps you are not sure if a component can be sourced that can achieve a specific design purpose, such as a heated powder bed for the SLM machine. Example solution: explore an in-house solution, consider local options and/or hiring a device.
 - Only include **safety risks if it could be an IMPEDIMENT to your project**. For instance, Ti6Al4V powder is flammable and if safety procedures are not in place to safely handle the powder then nickel powders should rather be used. Or, create comprehensive safety procedures for safe handling of Ti-powder according to relevant OHSA standards.

Conclusions

- Just as important as Introduction
- Not just a quick summary
- Answers this question:
 - Did you achieve what you set out to do (objectives and aim) and how?
- Every report should have one
- Not extremely vital in project proposal but vital in the rest of the drafts
- An example of a project proposal conclusion:
 - “Aerospace designs are increasingly more challenging to manufacture as efficiency and performance requirements are pushed to new limits. SLM offers a suitable manufacturing route for complex blade designs although special attention should be offered to resulting mechanical properties. The results of this project will allow recommendations to be made regarding optimum manufacturing parameters that maximise part density and mechanical behaviour. The project is expected to take 11 months to complete (concluding 23 October 2026) within a budget of R340 000. Running and capital costs are expected to be within the allocated R5000 budget. Risks related to safely handling powders will be mitigated through the use of OHSA standardised safety protocol and suitable equipment. “

Appendices: ECSA Self-Assessment (final)

ECSA Graduate Attributes



Problem Solving
The engineering problem is analysed and defined in chapter 1. The necessary engineering background and information is presented in chapter 2. Possible solutions and models are discussed and selected in chapters 2 (mechanical random sources), 3 (testing software selection), 4 (material and component selection) and 5 (software). The evaluation and conclusions are shown in chapters 7 (results) and 9 (conclusion), and show the system is a possible solution to the problem described in chapter 1.
Application of scientific and engineering knowledge
Mathematical and numerical analysis can be found in chapter 2 (derivation of equations of motion for double pendulum), 5 (power supply construction) and 7 (test results). Physical laws and knowledge is shown throughout the report (look to chapters 4, 5 and 6). Engineering principles are also displayed throughout the report (look to chapters 4, 5 and 7).
Engineering design
The planning and designing of the system including the different analyses and use of students own judgement is shown in chapters 2 (mechanical random sources) , 4 (alternatives), 5 (CAD and 3D printing) and 6 (software). The logical design flow is shown in chapter 3. The results provide significant practical value to other similar developments.
Engineering methods, skills and tools, including information technology
Modules from 3rd and 4th year engineering such as e-design and introduction to systems engineering is used in this project as seen in chapters 3, 4, 5 and 6. Use of software is shown in chapter 5 where CAD was used to design the physical model and complex software such as the National Institute of Standards and Technology's test suite for randomness is used in chapter 7.
Professional and technical communication
Project proposal, progress report, final draft and final report show the style and language used to convey the problem and solution. The video and oral presentation provide graphical support and professional communication. The report is created in a logical order starting with research, design, construction, and lastly testing.
Individual, team and multidisciplinary working
All aspects of the project. The use of visual graphics was used to enhance and explain the results of the system to provide further understanding and insight into the results.
Independent learning ability
Chapter 2 shows the ability of the student to conduct his own research and chapter 7 shows knowledge and testing outside the scope learnt in engineering. All are provided throughout chapters 2 to 7 and the two pages of references show the wide range of topics read by the student.

ECSA Graduate Attributes



ECSA outcome	Addressed in section:
ELO 1. Problem solving: Demonstrate competence to identify, assess, formulate and solve convergent and divergent engineering problems creatively and innovatively.	3, 4, 5, 6
ELO 2. Application of scientific and engineering knowledge: Demonstrate competence to apply knowledge of mathematics, basic science and engineering sciences from first principles to solve engineering problems.	4, 5, 6, Appendix A2
ELO 3. Engineering design: Demonstrate competence to perform creative, procedural and non-procedural design and synthesis of components, systems, engineering works, products or processes.	3, Appendix A1
ELO 5. Engineering methods, skills and tools: Demonstrate competence to use appropriate engineering methods, skills and tools, including those based on information technology.	5, 6, Appendix A1, A2, B1 and C1
ELO 6. Professional and technical communication: Demonstrate competence to communicate effectively, both orally and in writing, with engineering audiences and the community at large.	1, 2, 7, progress report, progress presentation, poster and final report presentation,
ELO 8. Individual, team and multidisciplinary working: Demonstrate competence to work effectively as on individual, in teams and in multidisciplinary environments.	2, 3, 4, 5, 6, meeting minutes
ELO 9. Independent learning ability: Demonstrate competence to engage in independent learning through well-developed learning skills.	2, 3, 4, 5, 6

Appendices: ECSA Self-Assessment (proposal)

- For the PROJECT PROPOSAL ONLY (because you have no “final report” yet)
- Typically, one page
- Indicate during what stages/activities of your project will you demonstrate the ECSA graduate attributes
- E.g.

ECSA graduate attribute	Project Activities
Engineering design	During the tensile specimen design phase (Activity 5) and the experimental methodology development for post-print testing (Activity 4)
Engineering methods, skills and tools (including information technology)	Use of computer-aided design (CAD) software packages, Matlab or other mathematical programming suites for statistical processing, use of Digital Image Correlation softwares and equipment for strain measurement during tensile testing (Activity 7)
Independent learning ability	Research on optimum processing parameters and suitable powders/printable designs in the literature review stage (Activity 1)

Appendices: Resource Use and End-Of-Life Strategy

- New (since 2022)! Please attend lecture and see guideline on STEMLearn
- Example below (BUT FOLLOW GUIDELINES ON STEMLEARN FIRST AND FOREMOST)

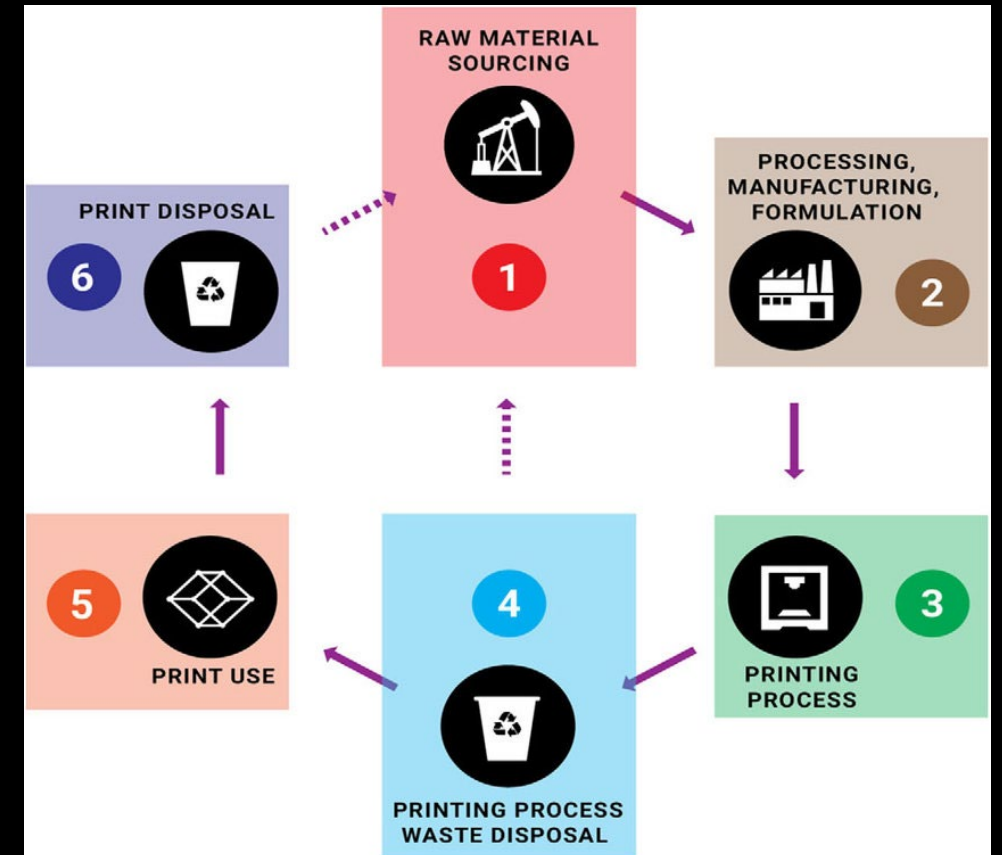
Life cycle stage no.	Life cycle stage	Level 2 summary	Level 1 summary	Level 0 summary
3	Printing process	Consumer-friendly material; can be used in an environment with untrained users	Professional use only; printing process introduces some hazards that require training or hazard controls.	Highly hazardous resins; AM materials/processes require additional safety controls.
4	Printing process waste disposal	No waste or any waste produced can be reused or biodegrades readily.	Some material waste; landfillable in most countries	Some material waste; uncured material waste needs to be disposed of as hazardous waste.
5	Print use	Safe for long-term contact (oral and dermal) (greater than 24 h)	Safe for intermittent skin contact (greater than 1 h to less than 24 h)	Unsafe for more than temporal skin contact (<1 h)
6	Print disposal	Readily biodegradable or recyclable	Landfillable in most countries	Material is hazardous waste; degrades into toxic chemicals; absorbs hazardous chemicals

Note: AM = additive manufacturing; h = hour(s).

Source: Bours et al. (2017)

E.g. strategy:

Reuse powders, recycle machine offcuts, use recyclable materials, repurpose printer base plate for future prints



Source: Bours et al. (2017)

To summarise: Main elements

- Executive summary
- ECSA graduate attribute self-assessment (for proposal can also be in Appendix)
- Introduction
- Project aim and objectives (include a scope if you must)
- Motivation
- Planned activities and risks
- Timeline (Gantt chart)
- Budget
- Appendix: Responsible use of resources and end-of-life strategy
- Reference list (Harvard referencing preferred – Name, date).